Chapter 3 Duolingo Math checklist

Tick each textbook section as you read it and each Duolingo unit as you finish it. Units are listed per lane (a Duolingo grade or topic) in the order the app presents them, so you can follow one lane top to bottom. Ratings: full = covers the section's main skill, partial = covers part of it, prereq = a warm-up from an earlier grade. The right column names the textbook sections the unit serves.

Chapter 3: Exponential and Logarithmic Functions (p. 215)

Grade 11 covers exponentials and logarithms almost section by section. Duolingo stops at growth and decay, so the Gaussian, logistic, and logarithmic models of 3.5 come from the book only.

Textbook sections

☐	3.1 Exponential Functions and Their Graphs	p. 216
☐	3.2 Logarithmic Functions and Their Graphs	p. 227
☐	3.3 Properties of Logarithms	p. 237
☐	3.4 Exponential and Logarithmic Equations	p. 244
☐	3.5 Exponential and Logarithmic Models	p. 255

Book only, not on Duolingo: **3.1** Compound interest, n times per year and continuously. **3.4** Logarithmic equations; Checking for extraneous solutions. **3.5** Gaussian models; Logistic growth models; Logarithmic models (pH, Richter scale, decibels).

Grade 9

☐	70. Exponential function tables	prereq	3.1
☐	71. Exponential function graphs	prereq	3.1
☐	72. Exponential function equations	prereq	3.1

Grade 11

☐	38. Exponential function tables	full	3.1
☐	39. Exponential function graphs	full	3.1
☐	40. Exponential function equations	full	3.1
☐	41. Exponential growth	full	3.1, 3.5
☐	42. Exponential decay	full	3.1, 3.5
☐	48. Intro to the natural base	full	3.1
☐	49. Continuous growth models	full	3.1, 3.5
☐	50. Intro to logarithms	full	3.2
☐	51. Logarithm value	full	3.2
☐	52. Product rule	full	3.3
☐	53. Quotient rule	full	3.3
☐	54. Power rule	full	3.3
☐	55. Change of base formula	full	3.3
☐	56. Intro to logarithmic functions	full	3.2
☐	57. Logarithmic functions in base 10	full	3.2
☐	58. Logarithmic functions in base 2	full	3.2
☐	59. Logarithmic function graphs	full	3.2
☐	60. Logarithmic function intercepts	full	3.2
☐	61. The natural logarithm	full	3.2
☐	62. Natural logarithm graphs	full	3.2
☐	63. Solve exponential equations using logarithms	full	3.4
☐	64. Solve base e equations using logarithms	full	3.4

Grade 12

☐	15. Properties of logarithms	full	3.3
☐	16. Undo exponentials	partial	3.2, 3.4

Algebra

☐	18. Logarithms	full	3.2
☐	19. Product and quotient rules	full	3.3
☐	20. Properties of logarithms	full	3.3
☐	21. Solve logarithmic and exponential equations	full	3.4

Algebraic Graphing

☐	15. Intro to exponential functions	full	3.1
☐	16. Exponential graphs and equations	full	3.1
☐	18. Exponential growth and decay	partial	3.5
☐	19. The natural base, e	full	3.1
☐	20. Logarithmic functions	full	3.2
☐	21. Logarithmic function graphs	full	3.2
☐	22. Natural logarithmic function	full	3.2

Section guide

For each section: the Duolingo units that practise it (already listed above with checkboxes), then the parts only the textbook teaches. Tick each book-only item once you have worked it in the book.

3.1 Exponential Functions and Their Graphs (p. 216)

Grades 9 and 11 both cover exponential functions; grade 11 adds the natural base and continuous growth. Compound interest formulas are book only.

Duolingo, in app order:
Grade 9: 70. Exponential function tables (prereq) · 71. Exponential function graphs (prereq) · 72. Exponential function equations (prereq)
Grade 11: 38. Exponential function tables (full) · 39. Exponential function graphs (full) · 40. Exponential function equations (full) · 41. Exponential growth (full) · 42. Exponential decay (full) · 48. Intro to the natural base (full) · 49. Continuous growth models (full)
Algebraic Graphing: 15. Intro to exponential functions (full) · 16. Exponential graphs and equations (full) · 19. The natural base, e (full)

Book only:

- [ ] Compound interest, n times per year and continuously

3.2 Logarithmic Functions and Their Graphs (p. 227)

Grade 11 covers logarithms, their graphs, and the natural logarithm nearly one to one with the section.

Duolingo, in app order:
Grade 11: 50. Intro to logarithms (full) · 51. Logarithm value (full) · 56. Intro to logarithmic functions (full) · 57. Logarithmic functions in base 10 (full) · 58. Logarithmic functions in base 2 (full) · 59. Logarithmic function graphs (full) · 60. Logarithmic function intercepts (full) · 61. The natural logarithm (full) · 62. Natural logarithm graphs (full)
Grade 12: 16. Undo exponentials (partial)
Algebra: 18. Logarithms (full)
Algebraic Graphing: 20. Logarithmic functions (full) · 21. Logarithmic function graphs (full) · 22. Natural logarithmic function (full)

3.3 Properties of Logarithms (p. 237)

Fully covered: product, quotient, and power rules plus change of base in grade 11, consolidated in grade 12.

Duolingo, in app order:
Grade 11: 52. Product rule (full) · 53. Quotient rule (full) · 54. Power rule (full) · 55. Change of base formula (full)
Grade 12: 15. Properties of logarithms (full)
Algebra: 19. Product and quotient rules (full) · 20. Properties of logarithms (full)

3.4 Exponential and Logarithmic Equations (p. 244)

Exponential equations are covered well; logarithmic equations and checking for extraneous solutions are lighter in the app.

Duolingo, in app order:
Grade 11: 63. Solve exponential equations using logarithms (full) · 64. Solve base e equations using logarithms (full)
Grade 12: 16. Undo exponentials (prereq)
Algebra: 21. Solve logarithmic and exponential equations (full)

Book only:

- [ ] Logarithmic equations
- [ ] Checking for extraneous solutions

3.5 Exponential and Logarithmic Models (p. 255)

Only exponential growth and decay models are covered. Gaussian, logistic, and logarithmic models are book only.

Duolingo, in app order:
Grade 11: 41. Exponential growth (partial) · 42. Exponential decay (partial) · 49. Continuous growth models (partial)
Algebraic Graphing: 18. Exponential growth and decay (partial)

Book only:

- [ ] Gaussian models
- [ ] Logistic growth models
- [ ] Logarithmic models (pH, Richter scale, decibels)

Textbook: Ron Larson and Robert P. Hostetler. Precalculus with Limits, 2nd ed. (Pre-AP National Student Edition) Thomson Brooks/Cole, 2009. ISBN 978-0-538-73659-6. Duolingo inventory: 820 units, last verified 2026-10-07 on iOS. Only titles and unit names are reproduced, for reference; no textbook or Duolingo content is included. This checklist is licensed CC BY 4.0; see the repository NOTICE for third-party material.